

	current I in the circuit increases. Using $V = E - Ir$, the terminal p.d. V decreases.
6(a)	There are currently no massive bodies, such as black holes and neutron stars in the Solar System, capable of emitting gravitational waves which are detectable.
6(b)(i)	From Fig. 6.2, using data from LIGO Washington, Max fraction change in length, $\frac{\Delta L_{\max}}{L} = 1.2 \times 10^{-21}$ $\Delta L_{\max} = 1.2 \times 10^{-21} L$ $= 1.2 \times 10^{-21} (4.0 \times 10^3)$ $= 4.8 \times 10^{-18} \text{ m}$
6(b)(ii)	From (b)(i) , the maximum change of length of one of the vacuum tubes in the LIGO detector is very small. The path length difference between the two light beams will be very small if they are reflected once. By reflecting the two light beams many times, the path length difference, and hence the phase difference between them upon recombination will be sufficiently large for it to be detected.
6(b)(iii)	Theoretically, the two laser light beams have a phase difference of π radians. In the absence of gravitational waves, the two tubes have the same length and the two beams have the same path length. Hence, the two beams will still be anti-phase upon recombination, and will undergo destructive interference to produce a signal of zero amplitude.
6(c)(i)	Energy of each photon, $E = \frac{hc}{\lambda}$ In one second, number of photons N passing a point in the laser beam will have energy $= NE$. Energy passing per unit time is power P. So $NE = P$ $\rightarrow N = \frac{P}{E} = \frac{P}{\frac{hc}{\lambda}} = \frac{P\lambda}{hc}$

No.	Solution
	$= \frac{(100)(1064 \times 10^{-9})}{(6.63 \times 10^{-34})(3.00 \times 10^8)}$ $= 5.349 \times 10^{20}$ $\approx 5.35 \times 10^{20} \text{ s}^{-1}$
6(c)(ii)	When the laser beam is completely reflected back along its original path, each photon experiences a change in momentum of $2p$, where $p = \frac{h}{\lambda}$ (by de Broglie Hypothesis). By Newton's law, the average force exerted by the light beam on one mirror, $F = \frac{\Delta p}{\Delta t} = N \times 2p = N \times 2 \frac{h}{\lambda}$ $= (5.349 \times 10^{20}) 2 \left(\frac{6.63 \times 10^{-34}}{1064 \times 10^{-9}} \right)$ $= 6.666 \times 10^{-7}$ $\approx 6.67 \times 10^{-7} \text{ N}$
6(c)(iii)	When a photon is absorbed by the mirror, the photon experiences a change in momentum of p, where $p = \frac{h}{\lambda}$ (by de Broglie Hypothesis), instead of $2p$. Since force is directly proportional to rate of change of momentum by Newton's second law, the average force exerted by the light beam on one mirror will be reduced.
6(d)(i)	Using Einstein mass-energy equation, energy released, $E = \Delta mc^2$ $= (29 + 36 - 62)(2.0 \times 10^{30})(3.00 \times 10^8)^2$ $= 5.4 \times 10^{47} \text{ J}$
6(d)(ii)	Power per unit area received on Earth $= \frac{P}{4\pi r^2} = \frac{\frac{E}{t}}{4\pi r^2} = \frac{\frac{5.4 \times 10^{47}}{20 \times 10^{-3}}}{4\pi \times ((1.3 \times 10^9)(9.5 \times 10^{15}))^2}$ $= 1.409 \times 10^{-2}$ $\approx 1.4 \times 10^{-2} \text{ W m}^{-2}$
6(e)(i)	From Fig. 6.2, time between the detection of the gravitational waves at the two LIGO detectors = 0.012 s
6(e)(ii)	Speed of gravitational waves

No.	Solution
	$= \frac{\text{distance travelled}}{\text{time taken}} = \frac{3002 \times 10^3}{0.012}$ $= 2.502 \times 10^8$ $\approx 2.5 \times 10^8 \text{ m s}^{-1}$
6(e)(iii)	As the LIGO detectors are built on Earth, the gravitational waves will travel from free space into Earth, a denser medium, which is likely to cause the speed to decrease.
6(f)	1. Detectors in space are free from seismic and other forms of environmental vibrations. 2. The detectors need not have vacuum tubes as there is vacuum in space. The pair of reflecting mirrors in a detector may be placed much further apart for better accuracy of measurement.